

Object Detection

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DaSCI

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Readings

- Szeliski (2022), Chapter 6.3.
- Zhang, Lipton, Li and Smola (2023), *Dive into Deep Learning*, Chapter 14.3-14.8.
- Stanford University CS231n (2024): Deep Learning for Computer Vision. Lecture 9 ([Object Detection and Image Segmentation](#)).
- Liu et al. (2020). Deep learning for generic object detection: A survey. *International journal of computer vision*, 128, 261-318.
- Jiao et al. (2024). Multiscale Deep Learning for Detection and Recognition: A Comprehensive Survey. *IEEE Trans. on Neural Networks and Learning Systems*.

More classical approaches:

- Forsyth & Ponce (2012). Chapter 17 and 18.

Next slides are mainly based on those from [cs231n](#)

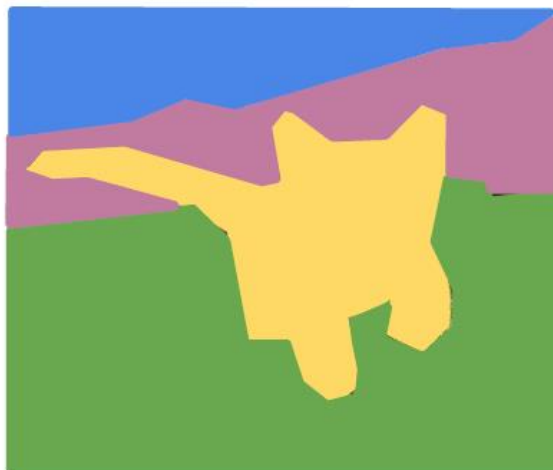
Classification



CAT

No spatial extent

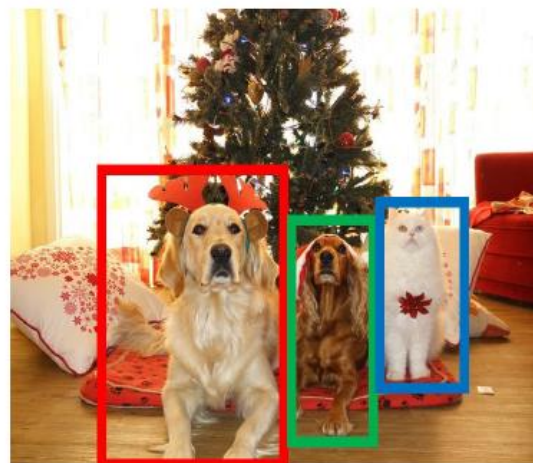
Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation

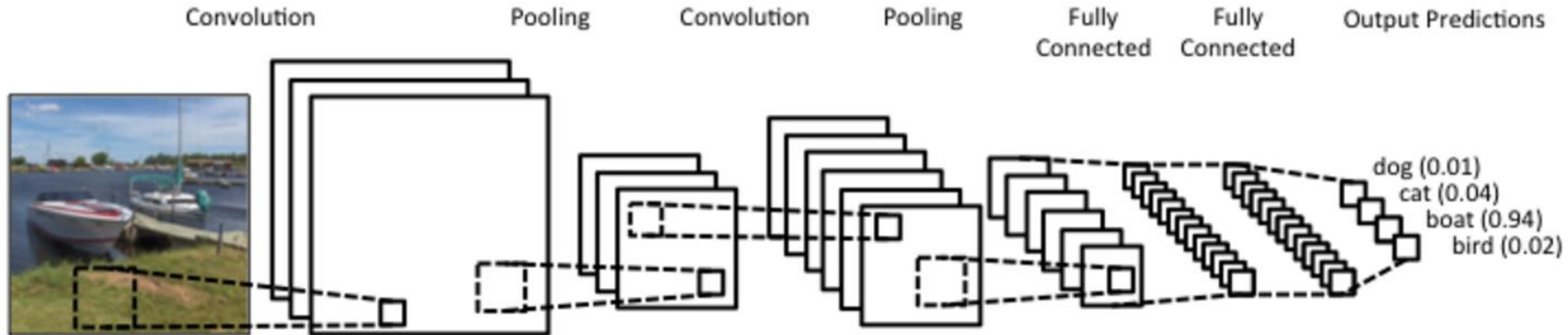


DOG, DOG, CAT

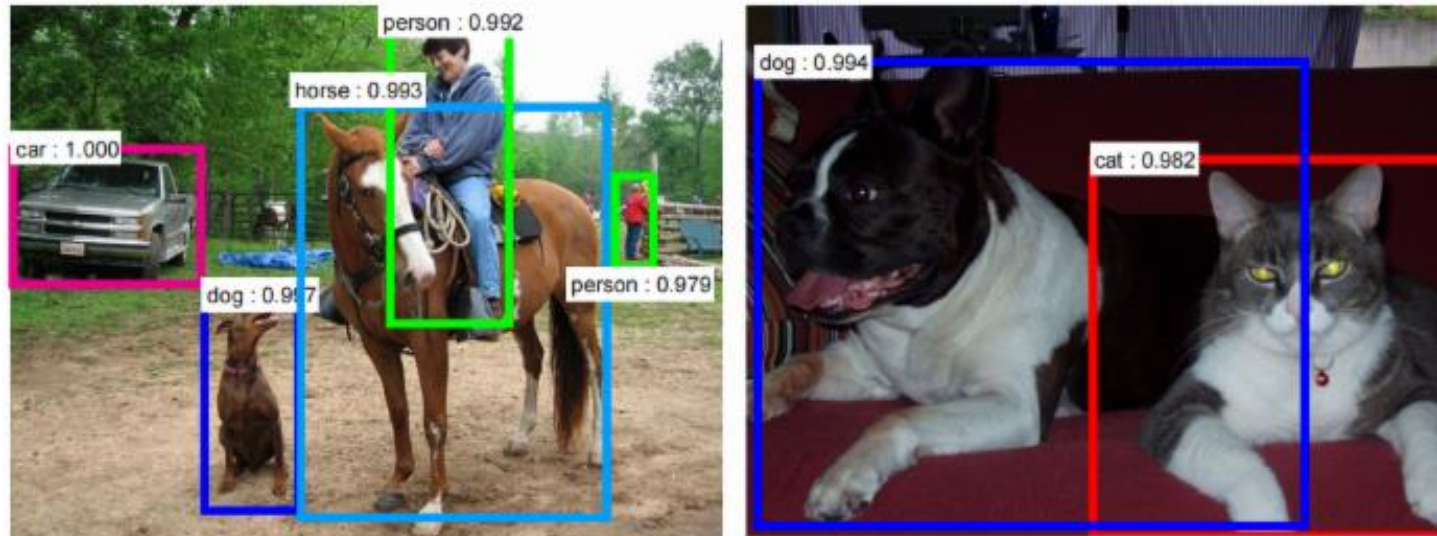
Multiple Object

From image classification to object detection

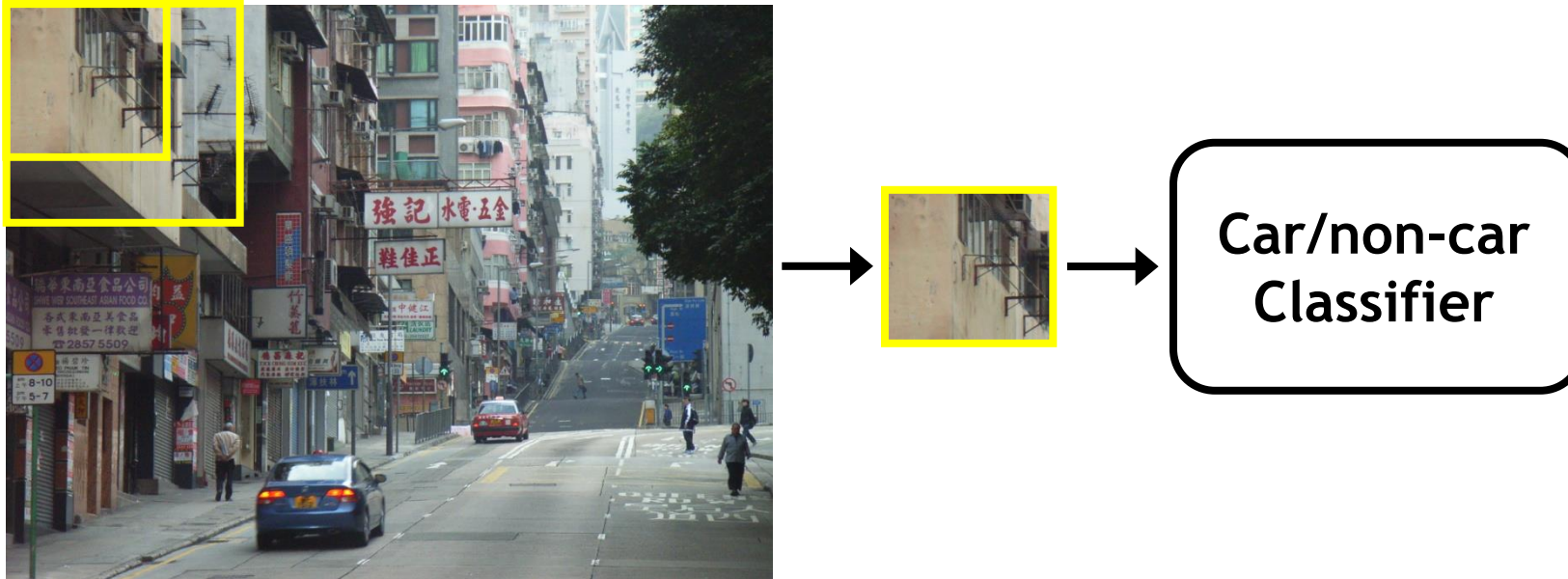
Image classification



Object detection



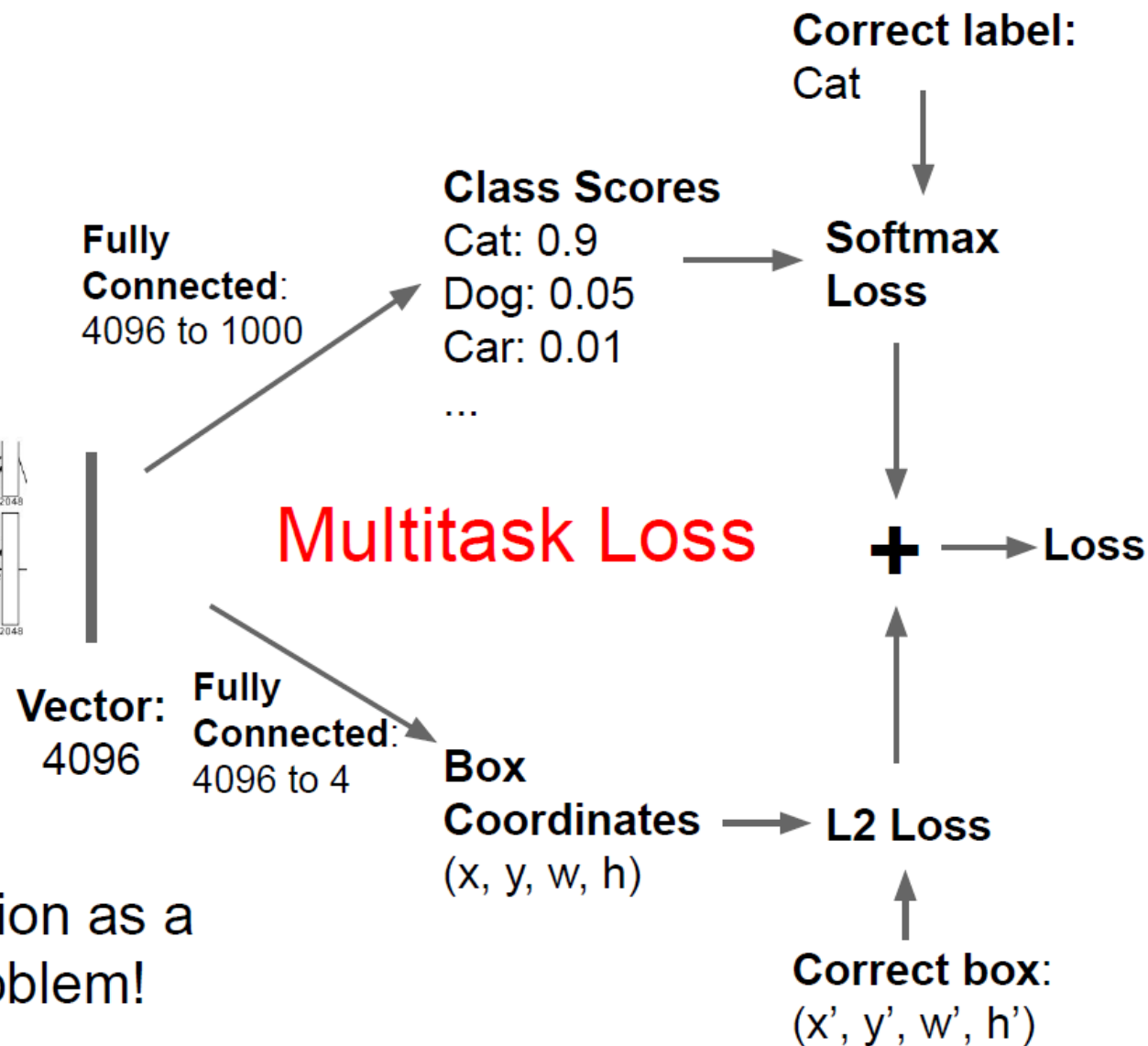
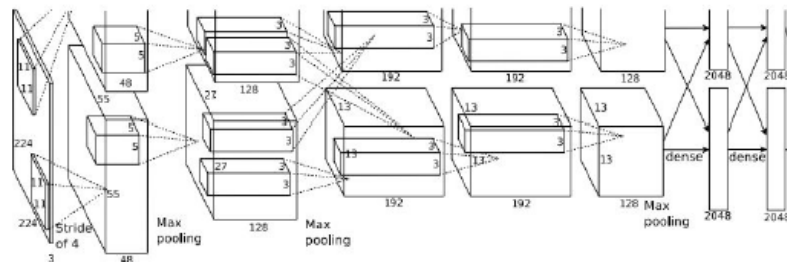
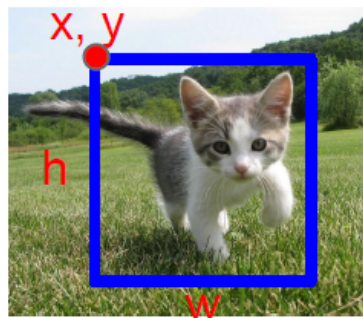
Sliding Window Detection



- Slide a window across the image and evaluate a detection model at each location
 - Thousands of windows to evaluate: **efficiency** and **low false positive rates** are essential
 - **Difficult to extend to a large range of scales, aspect ratios**

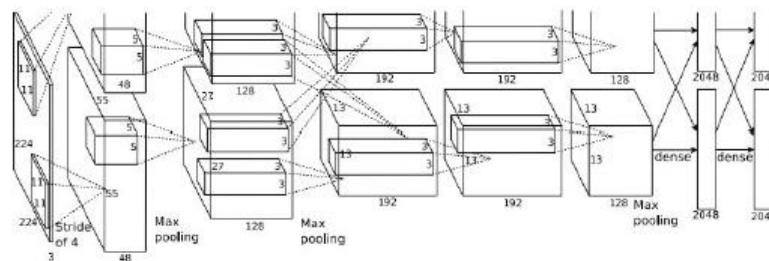
Object Detection: Single Object

(Classification + Localization)



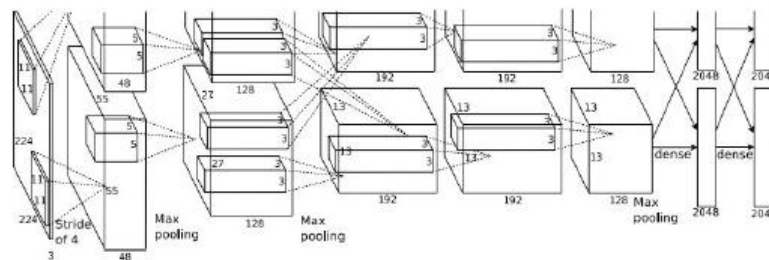
Object Detection: Multiple Objects

Each image needs a different number of outputs!



CAT: (x, y, w, h)

4 numbers

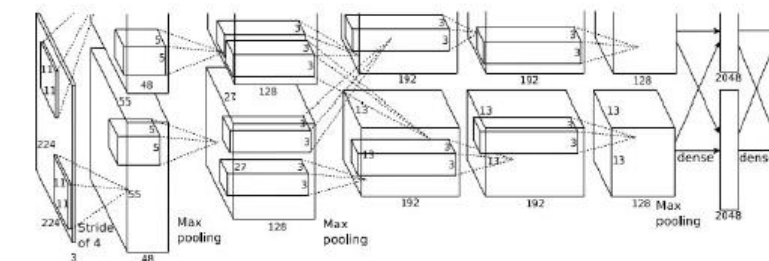


DOG: (x, y, w, h)

DOG: (x, y, w, h)

CAT: (x, y, w, h)

12 numbers



DUCK: (x, y, w, h)

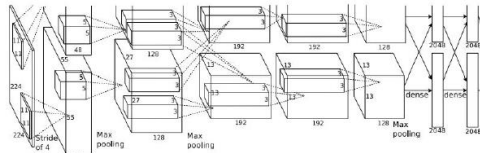
DUCK: (x, y, w, h)

....

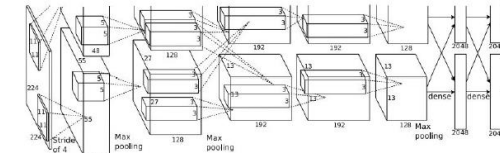
Many numbers!

Object Detection: Multiple Objects

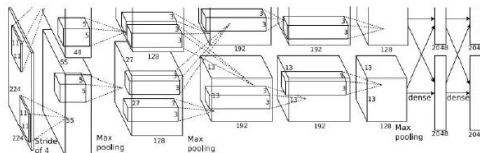
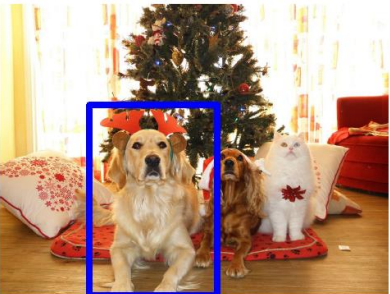
- Possible solution: Apply a ConvNet to many different image crops → ConvNet classifies each crop as object or background



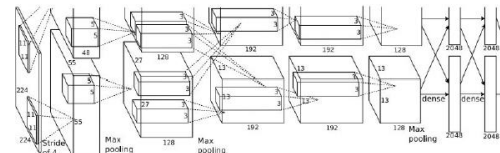
Dog? NO
Cat? NO
Background? YES



Dog? YES
Cat? NO
Background? NO



Dog? YES
Cat? NO
Background? NO



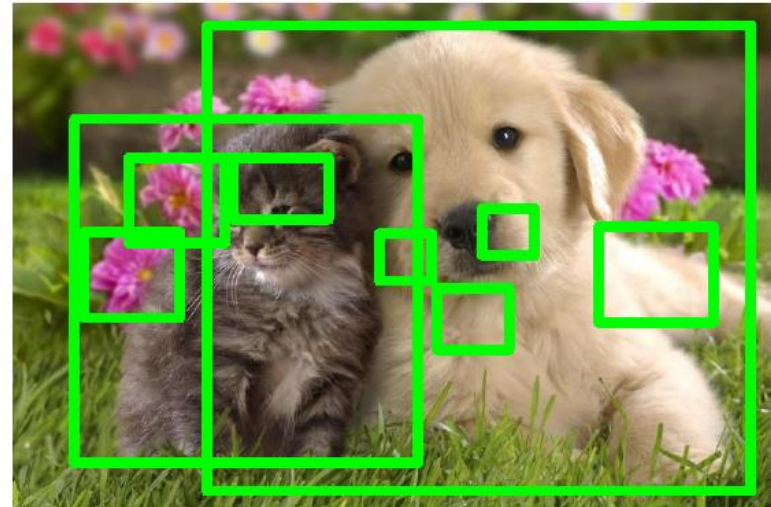
Dog? NO
Cat? YES
Background? NO

What's the problem with this approach?

Need to apply ConvNet to huge number of locations, scales, and aspect ratios, very computationally expensive!

Region Proposals: Selective Search

- Find “blobby” image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Alexe et al, “Measuring the objectness of image windows”, TPAMI 2012

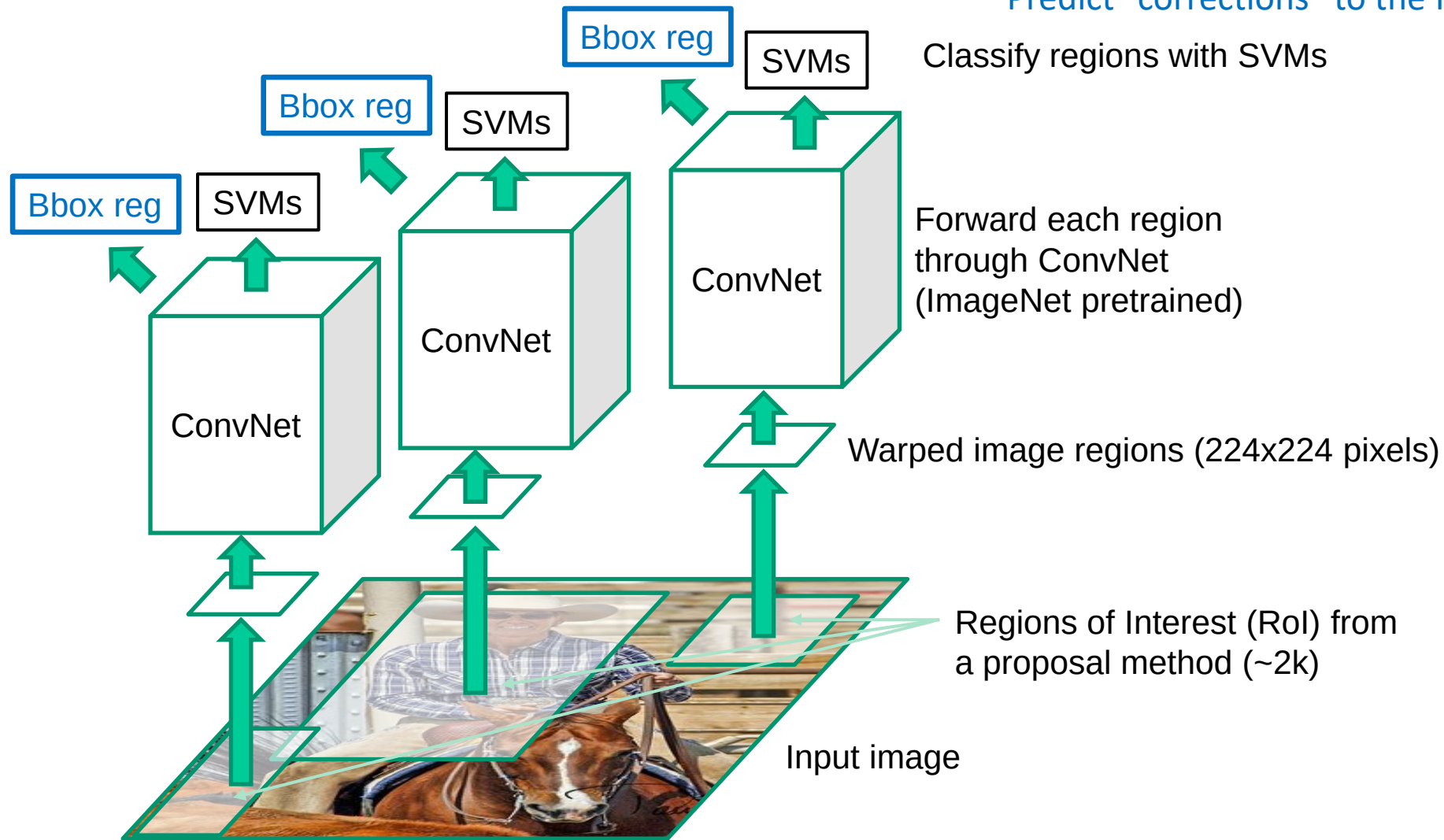
Uijlings et al, “Selective Search for Object Recognition”, IJCV 2013

Cheng et al, “BING: Binarized normed gradients for objectness estimation at 300fps”, CVPR 2014

Zitnick and Dollar, “Edge boxes: Locating object proposals from edges”, ECCV 2014

R-CNN

Predict “corrections” to the RoI: 4 numbers: (dx, dy, dw, dh)

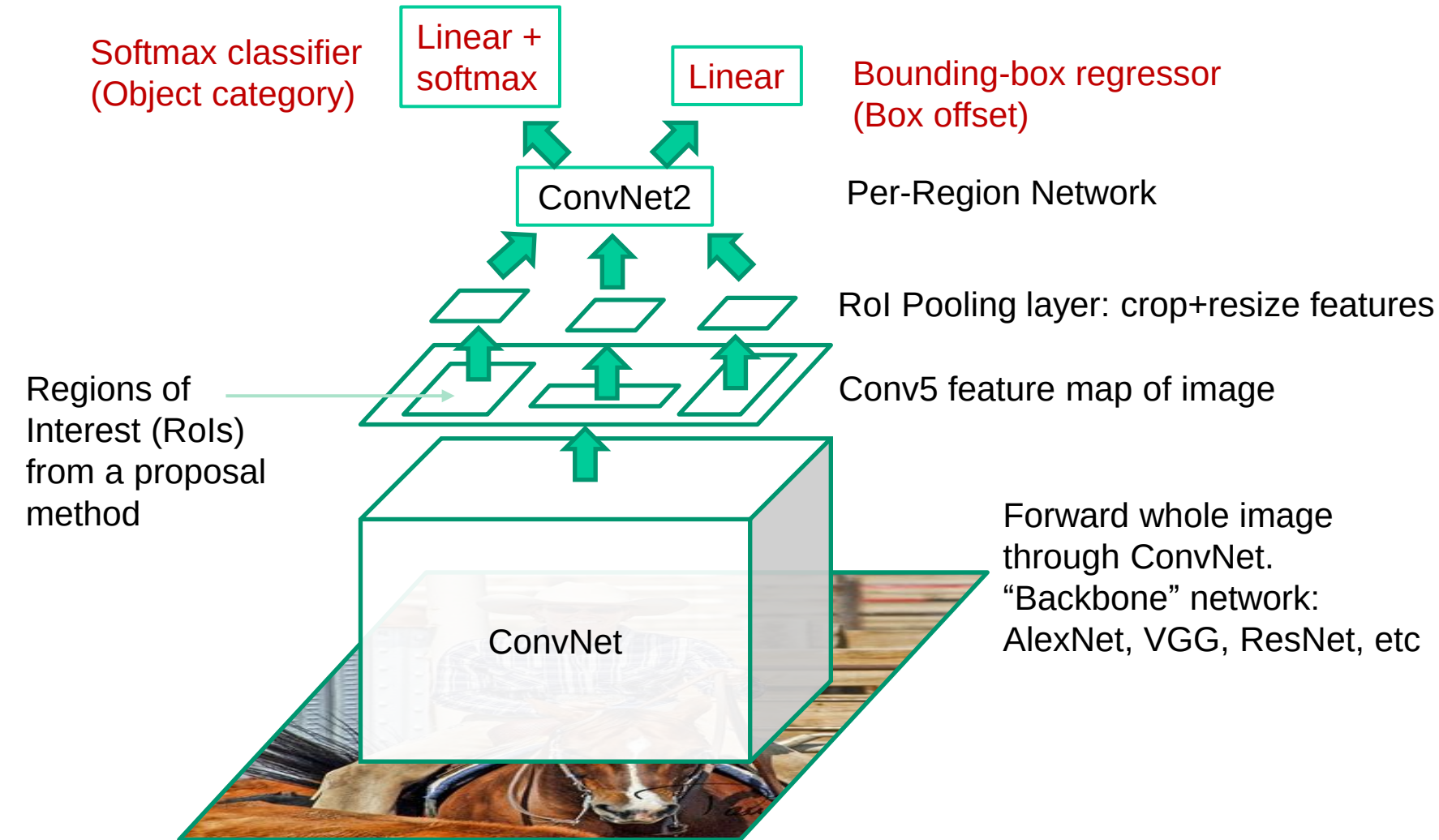


Problem:

- Very slow! Need to do ~2k independent forward passes for each image!
- Not a single end-to-end system!

Idea: Pass the image through ConvNet before cropping! Crop the conv feature instead!

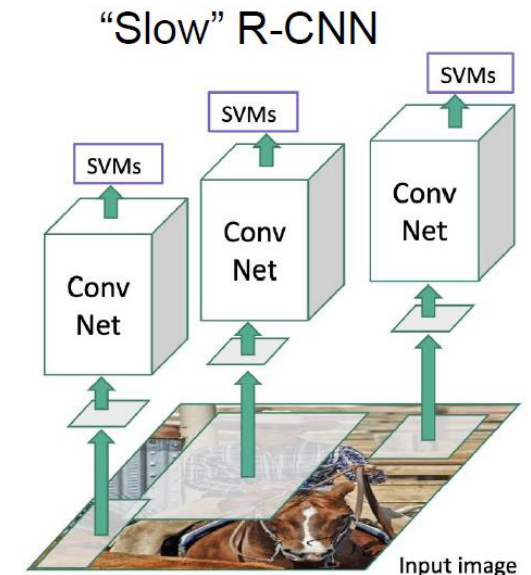
Fast R-CNN



Girshick et al., "Fast R-CNN", ICCV 2015.

Problem:
Runtime dominated
by region proposals!

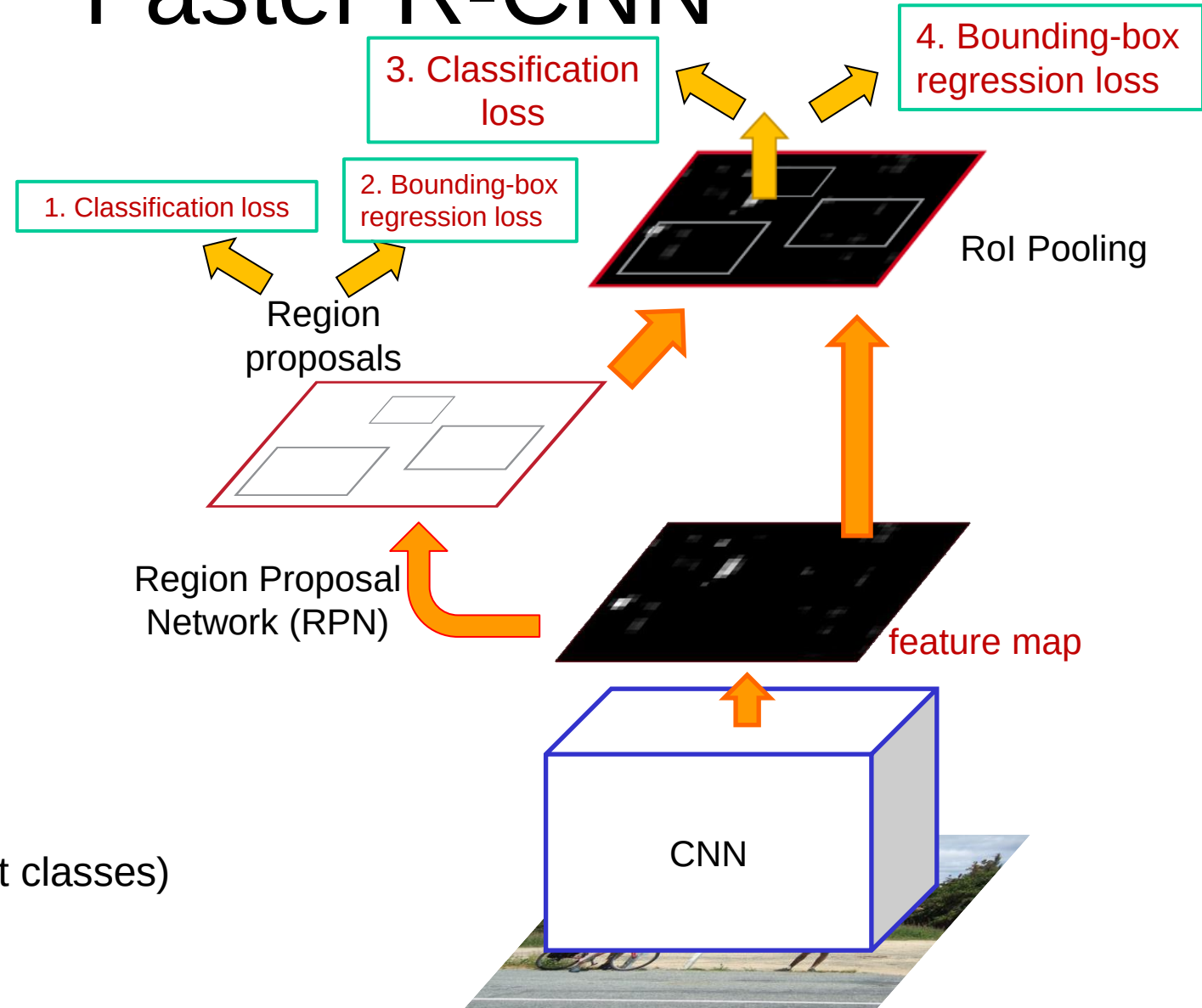
Idea: Make ConvNet
do proposals directly!



Faster R-CNN

Make ConvNet do proposals!

Insert **Region Proposal Network (RPN)** to predict proposals from features



Jointly train with 4 losses:

1. RPN classify object / not object
2. RPN regress box coordinates
3. Final classification score (object classes)
4. Final box coordinates

Region Proposal Network (RPN)

We use an **anchor box** (1x20x15) of fixed size at each point in the feature map

We use **K** different anchor boxes of different size / scale at each point



Input Image
(e.g. 3 x 640 x 480)

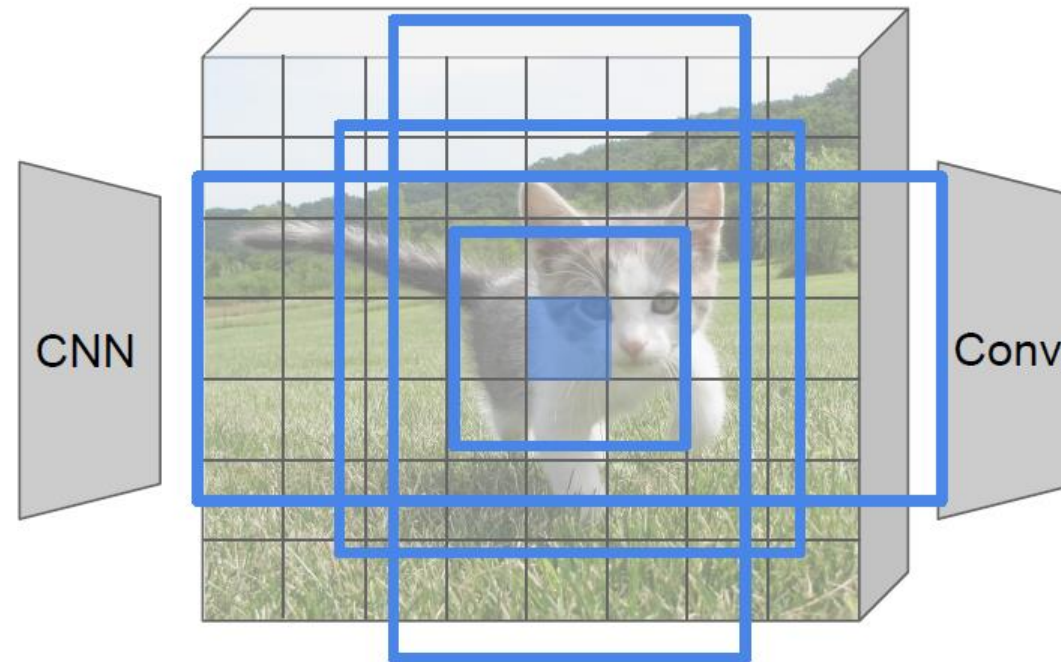


Image features
(e.g. 512 x 20 x 15)

At each point, **predict whether the corresponding anchor contains an object** (binary classification)

Anchor is an object?
 $K \times 20 \times 15$

Box transforms
 $4K \times 20 \times 15$

For positive boxes, also predict **corrections** from the anchor to the ground-truth box (regress 4 numbers per pixel)

We sort the $K \times 20 \times 15$ boxes by their “objectness” score, take top ~300 as our proposals

Faster R-CNN

Faster R-CNN is a
Two-stage object detector

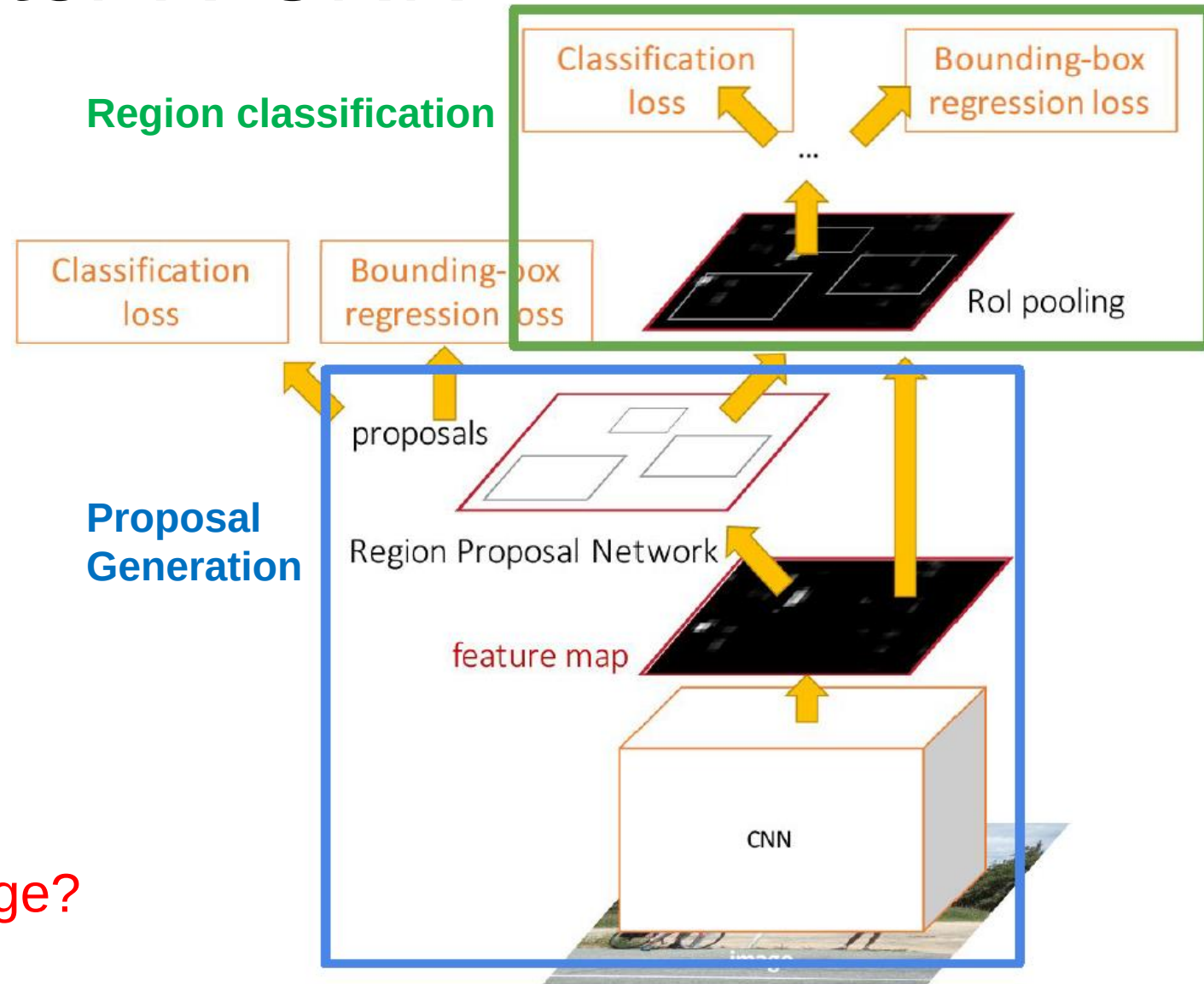
First stage: Run once per image

- Backbone network
- Region proposal network

Second stage: Run once per region

- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset

Do we really need the second stage?

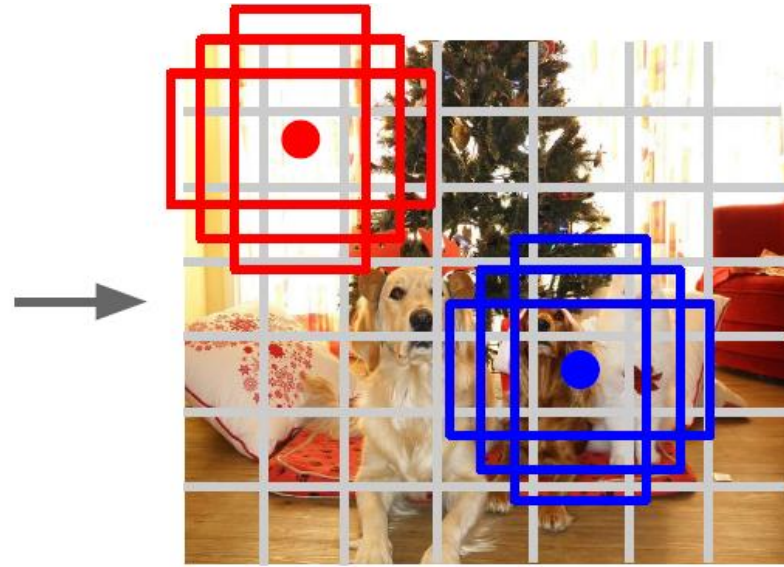


Single-Stage Object Detectors: YOLO / SSD / RetinaNet

In fact, YOLO means “You Only Look Once”...



Input image
 $3 \times H \times W$



Divide image into grid
 7×7

Image a set of **base boxes** centered
at each grid cell. Here $B = 3$

Within each grid cell:

- Regress from each of the **B** base boxes
5 numbers: **(dx, dy, dh, dw, confidence)**
- Predict **scores** for each of **C** classes
(including background as a class)
- **Looks a lot like RPN**, but category-specific!

Output: $7 \times 7 \times (5 * B + C)$ numbers

Redmon et al, “You Only Look Once: Unified, Real-Time Object Detection”, CVPR 2016
Liu et al, “SSD: Single-Shot MultiBox Detector”, ECCV 2016
Lin et al, “Focal Loss for Dense Object Detection”, ICCV 2017

Object Detection: Lots of Variables...

Backbone Network

VGG

ResNet

Inception

EfficientNet

...

“Meta-Architecture”

Two-stage: Faster R-CNN

Single-stage: YOLO / SSD

Methodological choices

- # Region Proposals
- How are anchors determined?
- How do we sample positive/negative samples for training the RPN?
- ...

Some of the main takeaways

Faster R-CNN is slower
but more accurate

SSD is much faster but
not as accurate

Deeper backbones work better

Huang et al, “Speed/accuracy trade-offs for modern convolutional object detectors”, CVPR 2017

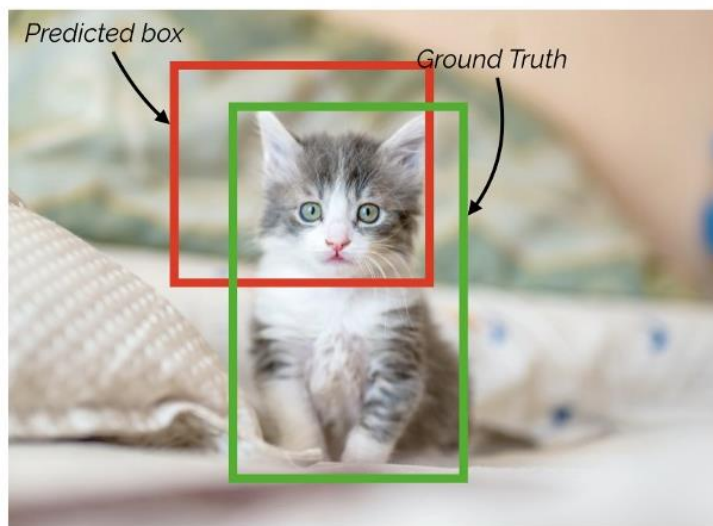
Dai et al, “R-FCN: Object Detection via Region-based Fully Convolutional Networks”, NIPS 2016

Object detection evaluation

- At test time, predict bounding boxes, class labels, and confidence scores
- For each detection, determine whether it is a TP or FP
 - Common criterion: $\text{Area}(\text{GT} \cap \text{Det}) / \text{Area}(\text{GT} \cup \text{Det}) > 0.5$
 - For multiple detections of the same GT box, only one considered as a TP

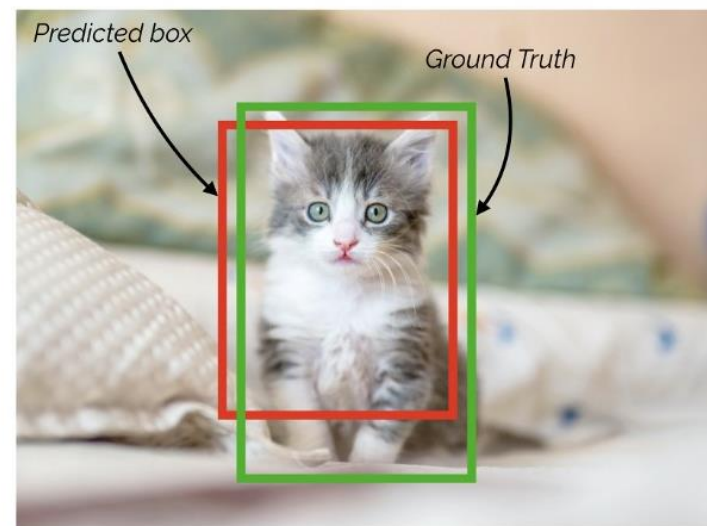
IoU threshold = 0.5

False Positive (FP)



IoU = ~0.3

True Positive (TP)



IoU = ~0.7

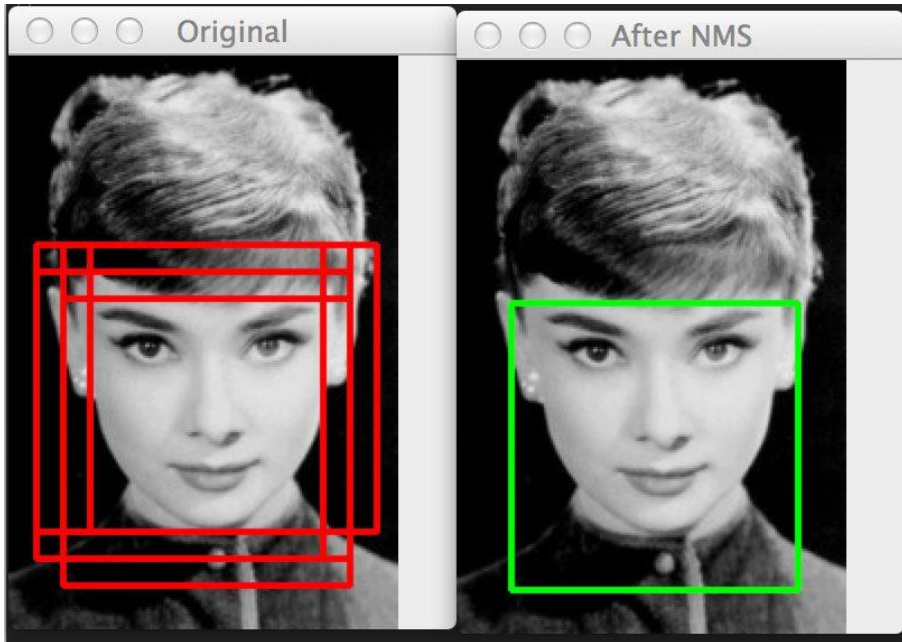
Non-Maxima Suppression (NMS)

What if same object is detected multiple times?

NMS eliminates some candidates that are in fact different detections of the same object.



We eliminate boxes that overlap significantly with a higher scoring bounding box



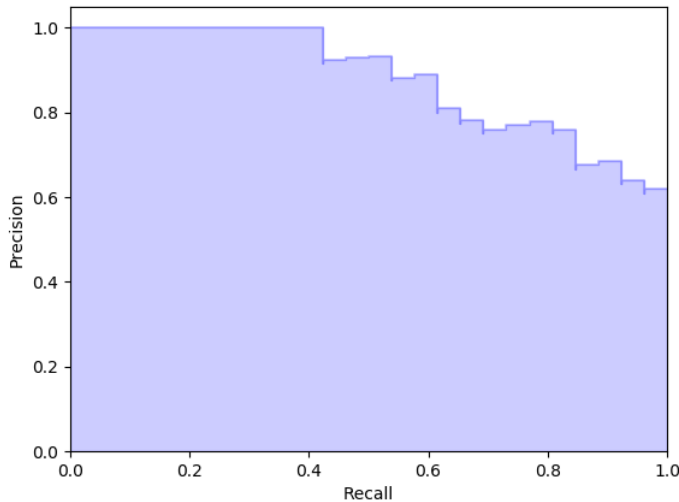
[Image Source](#)

For each object class c :

1. Discard all boxes with $p_c \leq 0.6$
2. While there are remaining boxes:
 - i. Keep the box with the largest p_c .
 - ii. Discard any remaining box with $\text{IoU} > 0.5$ with the box selected in (i)

Object detection evaluation

- For each class, plot **Recall-Precision curve** and compute **Average Precision** (area under the curve)
- Take mean of AP over classes to get **mAP**



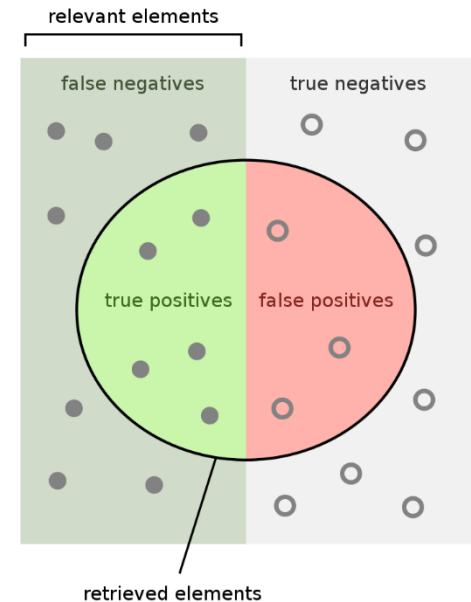
Precision:

true positive detections / total detections
 $TP / (TP+FP)$

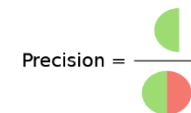
Recall:

true positive detections / total positive test instances
 $TP / (TP+FN)$

An object detector is considered good if its precision stays high as recall increases, which means that **if you vary the confidence threshold, the precision and recall will still be high.**



How many retrieved items are relevant?



Precision =

How many relevant items are retrieved?



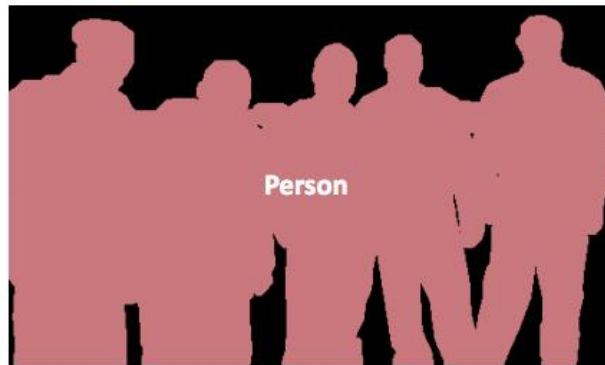
Recall =

Instance segmentation: Mask R-CNN

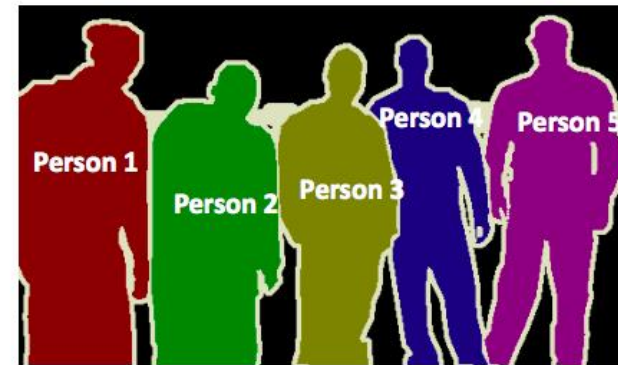
Instance Segmentation: **image segmentation distinguishing between different objects/instances of the same class**



Object Detection



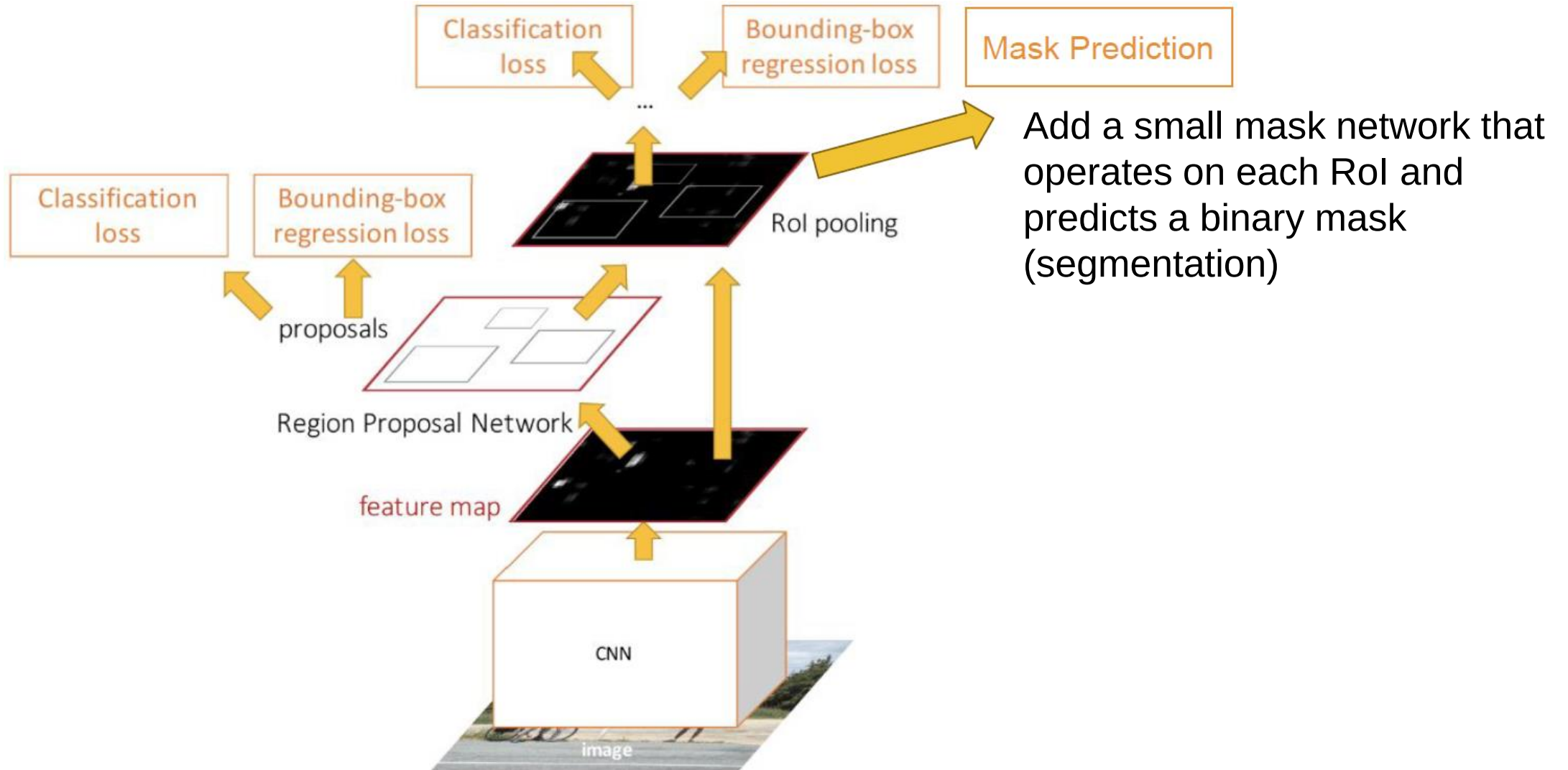
Semantic Segmentation



Instance Segmentation

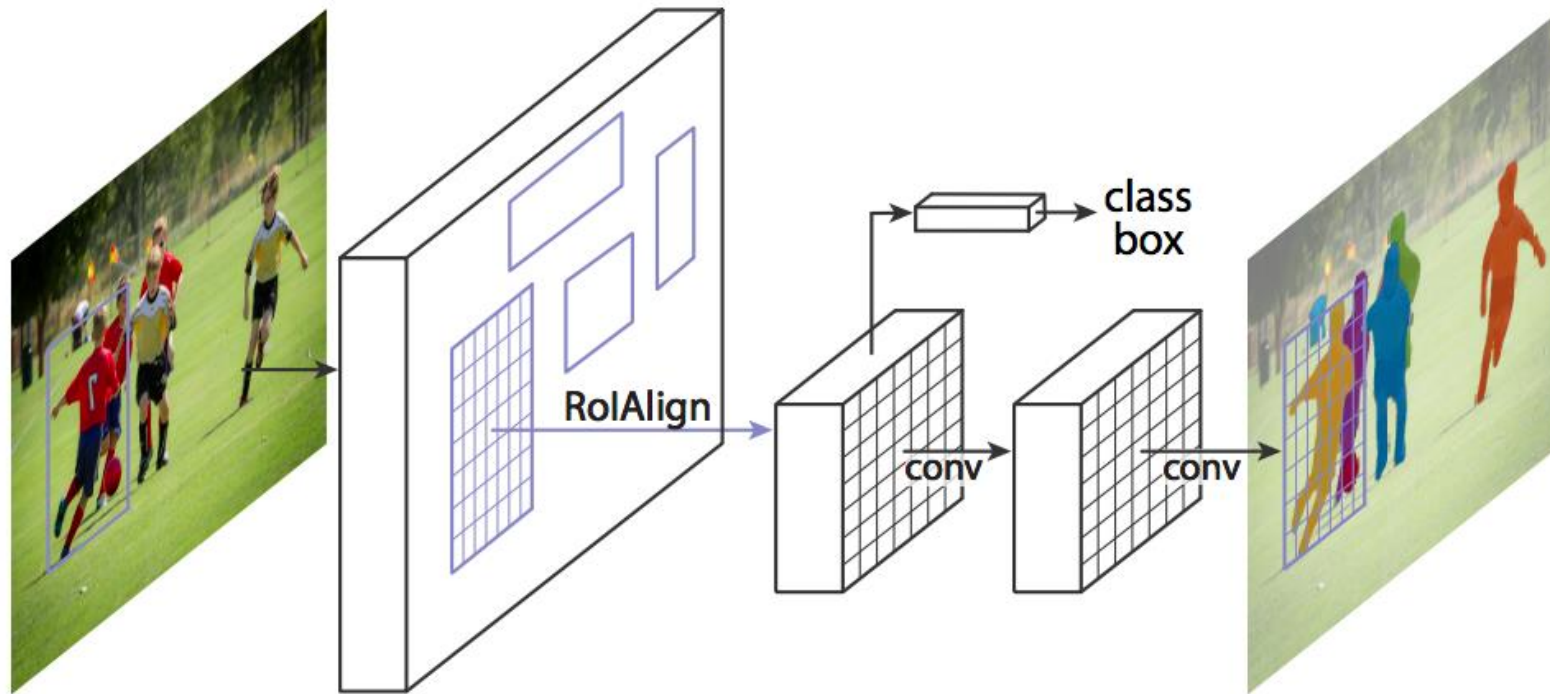


Instance segmentation: Mask R-CNN



Instance segmentation: Mask R-CNN

Mask R-CNN = Faster R-CNN + FCN on RoIs



Mask branch: separately predict segmentation for each possible class

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